Chapter 24

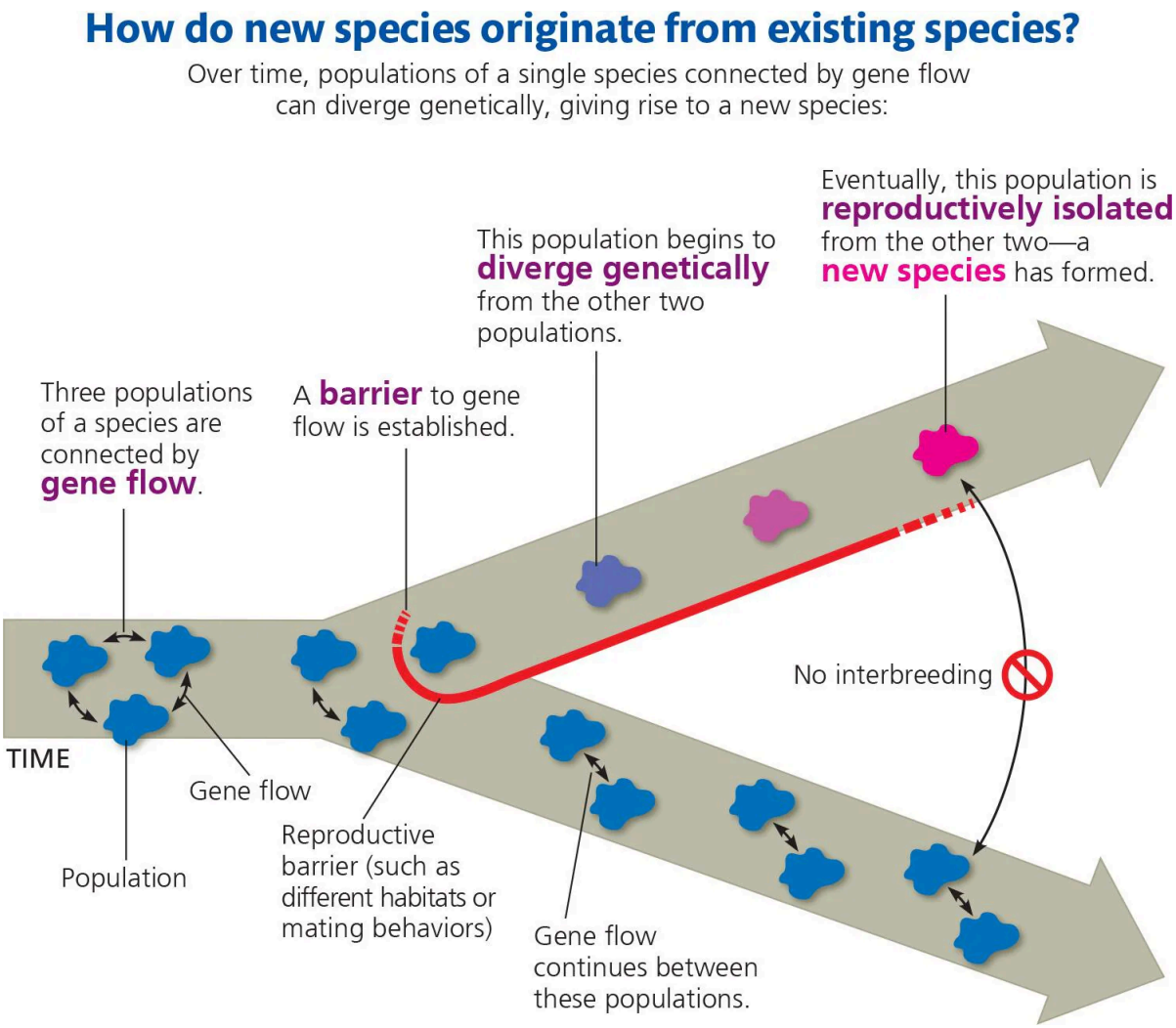
Campbell Biology, AP Edition, 12e

Chapter 24

The Origin of Species

Figure 24.1

This flightless cormorant is one of many species in the Galápagos Islands that are found nowhere else in the world. How did a bird that cannot fly come to live in this isolated place? When Darwin visited the Galápagos, he also was intrigued by its unique species, and later concluded that they must have originated in the islands from ancestors that had traveled from South America.



Over time, populations of a single species connected by gene flow can diverge genetically, giving rise to a new species. Begin with populations with gene flow. Three populations of a species are connected this way. A barrier to gene flow is established, such as a reproductive barrier (including different habitats or mating behaviors). One population begins to diverge genetically from the other two populations. Gene flow continues between the other two populations. Eventually, the lone population is reproductively isolated from the other two and a new species has formed.

Key Concepts

- **24.1The biological species concept emphasizes reproductive isolation****
- **24.2Speciation can take place with or without geographic separation****
- **24.3Hybrid zones reveal factors that cause reproductive isolation****
- **24.4Speciation can occur rapidly or slowly and can result from changes in few or many genes****

Speciation may occur when two populations become reproductively isolated from one another **(BIG IDEA 1)**. This can lead to differences in timing and coordination of behavior and changes in modes of intercellular and intracellular signal transmission **(BIG IDEA 3)**. Reproductive isolation can also affect relationships with other species in the environment and bring about adaptation or morphological change **(BIG IDEA 4)**.

Study Tip

Make a table: Some processes that can lead to speciation only occur in allopatric (geographically separated) populations, while others also occur in sympatric (geographically overlapping) populations. To help you keep track of these processes and the geographic conditions under which they can occur, fill in the table as you read the chapter.

Process (include page # or figure #)	Can occur in allopatric populations (yes/no)?	Can occur in sympatric populations (yes/no)?
Sexual selection (Figure 24.12)	Yes	Yes

The table is as follows:

Process (include page number and figure number)	Can occur in allopatric populations (yes or no)	Can occur in sympatric populations (yes or no)
Sexual selection (Figure 24.12)	Yes	Yes
blank	blank	blank

When Darwin came to the Galápagos Islands, he noted that they were teeming with unique species, like the flightless bird in Figure 24.1. Later, he realized that these species had formed relatively recently, writing in his diary, “Both in space and time, we seem to be brought somewhat near to that great fact—that mystery of mysteries—the first appearance of new beings on this Earth.”

The “mystery of mysteries” that captivated Darwin is **speciation**, the process by which one species splits into two species. Speciation fascinated Darwin (and many biologists since) because it has produced the tremendous diversity of life, repeatedly yielding new species that differ from existing ones. Speciation also helps to explain the many features that organisms share (the unity of life): When one species splits into two, the species that result share many characteristics because they are descended from this common ancestor. For example, at the DNA sequence level, such similarities indicate that the flightless cormorant in Figure 24.1 is closely related to flying cormorants found in the Americas. This suggests that the flightless cormorant originated from an ancestral cormorant species that flew from the mainland to the Galápagos.

Speciation forms a conceptual bridge between ****__microevolution__, changes over time in allele frequencies in a population, and **macroevolution**, the broad pattern of evolution above the species level. An example of macroevolutionary change is the origin of new groups of organisms, such as mammals or flowering plants, through a series of speciation events. We examined microevolutionary mechanisms in Chapter 23, and we'll turn to macroevolution in Chapter 25. In this chapter, we'll explore the mechanisms by which new species originate from existing ones. First, let's establish what we actually mean by a "species."

Get Ready for This Chapter

In this chapter, you will learn about mechanisms that can lead to speciation, the process by which one species splits into two or more species. Before beginning this chapter, you should be able to do the following:

- Distinguish meiosis from mitosis (see Figure 13.10).
- Explain the chromosomal basis of inheritance (see Figure 15.4 and Concept 15.1).
- Describe how mutation, gene flow, genetic drift, and natural selection affect allele frequencies in populations (see Concept 23.3).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following statements about meiosis is correct?

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In meiosis, each cell division produces two genetically identical daughter cel
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That's incorrect. Try again.

Incorrect. Review the following material: Meiosis in a diploid cell produces two haploid cells. In meiosis, the independent assortment of homologous chromosomes results in each daughter cell receiving one of 2n (where n is the haploid chromosome number) possible combinations of the maternal and paternal chromosomes. Crossing over may occur during meiosis, thereby producing recombinant chromosomes that carry genes derived from two different parents.

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Meiosis in a diploid cell produces diploid daughter cells.
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That's incorrect. Try again.

Incorrect. Review the following material: Meiosis in a diploid cell produces two haploid cells. In meiosis, the independent assortment of homologous chromosomes results in each daughter cell receiving one of 2n (where n is the haploid chromosome number) possible combinations of the maternal and paternal chromosomes. Crossing over may occur during meiosis, thereby producing recombinant chromosomes that carry genes derived from two different parents.

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Meiosis produces cells that differ genetically from the parent cell.
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That's correct.

Correct! If you had trouble with this question, review the following material: Meiosis in a diploid cell produces two haploid cells that differ genetically from one another and from the diploid parent cell. This occurs in part because in meiosis, the independent assortment of homologous chromosomes results in each daughter cell receiving one of 2n (where n is the haploid chromosome number) possible combinations of the maternal and paternal chromosomes. Another reason is that crossing over may occur during meiosis, thereby producing recombinant chromosomes that carry genes derived from two different parents.

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Crossing over cannot occur during meiosis.
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That's incorrect. Try again.

Incorrect. Review the following material: Meiosis in a diploid cell produces two haploid cells. In meiosis, the independent assortment of homologous chromosomes results in each daughter cell receiving one of 2n (where n is the haploid chromosome number) possible combinations of the maternal and paternal chromosomes. Crossing over may occur during meiosis, thereby producing recombinant chromosomes that carry genes derived from two different parents.

2. In a species that has two chromosomes, suppose that a male has genotype aa for a gene on chromosome 1 and genotype bb for a gene on chromosome 2. If a female of this species has genotype AA for the gene on chromosome 1 and genotype BB for the gene on chromosome 2, which of the following represents the possible genotypes in a cell of their offspring?

aB on chromosome 1, Ab on chromosome 2

That's incorrect. Try again.

Incorrect. Review the following material: Each sperm produced by the male contains one copy of chromosome 1 (carrying the a allele) and one copy of chromosome 2 (carrying the b allele), whereas each egg produced by the female contains one copy of chromosome 1 (carrying the A allele) and one copy of chromosome 2 (carrying the B allele).

AA on chromosome 1, bb on chromosome 2

That's incorrect. Try again.

Incorrect. Review the following material: Each sperm produced by the male contains one copy of chromosome 1 (carrying the a allele) and one copy of chromosome 2 (carrying the b allele), whereas each egg produced by the female contains one copy of chromosome 1 (carrying the A allele) and one copy of

chromosome 2 (carrying the B allele).

AB on chromosome 1, ab on chromosome 2

That's incorrect. Try again.

Incorrect. Review the following material: Each sperm produced by the male contains one copy of chromosome 1 (carrying the a allele) and one copy of chromosome 2 (carrying the b allele), whereas each egg produced by the female contains one copy of chromosome 1 (carrying the A allele) and one copy of chromosome 2 (carrying the B allele).

Aa on chromosome 1, Bb on chromosome 2

That's correct.

Correct! If you had trouble with this question, review the following material: Each sperm produced by the male contains one copy of chromosome 1 (carrying the a allele) and one copy of chromosome 2 (carrying the b allele), whereas each egg produced by the female contains one copy of chromosome 1 (carrying the A allele) and one copy of chromosome 2 (carrying the B allele). Thus, when any egg and sperm cells unite, the offspring cells will have genotype Aa on chromosome 1 and genotype Bb on chromosome 2.

3. A windstorm transports a few seeds of a plant species from Island A, where the species is abundant, to Island B, where prior to this event the plant species was not found. The environmental conditions on the two islands are very similar, but the islands are located far away from one another. Over time, a small population of this plant becomes established on Island B. A biologist samples plants on both islands and finds that allele frequencies on Island A differ substantially from allele frequencies on Island B at several genetic loci not known to affect survival or reproduction. Based on this information, the observed genetic changes mostly likely resulted from

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genetic drift.
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That's correct.

Correct! If you had trouble with this question, review the following material: Since, the environmental conditions on the two islands are very similar and the genetic loci sampled are not known to affect survival or reproduction, it is not likely that the observed differences were caused by natural selection. Gene flow causes populations to become more similar genetically over time, not more different. However, the observed differences could well be caused by genetic drift, the fluctuation of allele frequencies from one generation to the next in a population due to chance events; for example, the founder effect often causes members of a new population (such as that on Island B) to differ genetically from members of the source population (Island A).

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natural selection.
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That's incorrect. Try again.

Incorrect. Review the following material: Since the environmental conditions on the two islands are very similar and the genetic loci sampled are not known to affect survival or reproduction, it is not likely that the observed differences were caused by natural selection. Gene flow causes populations to become more similar genetically over time, not more different.

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mutation.
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That's incorrect. Try again.

Incorrect. Review the following material: Since the environmental conditions on the two islands are very similar and the genetic loci sampled are not known to affect survival or reproduction, it is not likely that the observed differences were caused by natural selection. Gene flow causes populations to become more similar genetically over time, not more different.

gene flow (after the species became established on Island B).

That's incorrect. Try again.

Incorrect. Review the following material: Since the environmental conditions on the two islands are very similar and the genetic loci sampled are not known to affect survival or reproduction, it is not likely that the observed differences were caused by natural selection. Gene flow causes populations to become more similar genetically over time, not more different.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 24.1Define the biological species concept, and identify reproductive barriers that could isolate a species' gene pool from that of other species.
- 24.2Compare and contrast allopatric speciation and sympatric speciation.
- 24.3Explain what a hybrid zone is, and identify possible outcomes for a hybrid zone over time.
- 24.4Give examples that show how speciation can occur rapidly or slowly, and can result from changes in few or many genes.